

# PRAVEEN T

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## EDUCATION

|                                                                         |                  |
|-------------------------------------------------------------------------|------------------|
| <b>Diplomo Engineering - Electrical and Electronics Engineering</b>     | <b>2021-2023</b> |
| Vellaichamy Nadar Polytechnic                                           | 88.8%            |
| <b>Bachelor of Engineering - Electrical and Electronics Engineering</b> | <b>2022-2026</b> |
| Chennai Institute of Technology                                         | CGPA-7.5         |

## PROFESSIONAL EXPERIENCE

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| <b>Designing Intern - DeSource Mobility</b>                                                                                                                                                                                                                                                                                                                                                 | 05/2025 – 05/2025 |
| <ul style="list-style-type: none"><li>• Gained hands-on experience in lithium-ion battery pack design, manufacturing, servicing, and safety practices, ensuring reliability and performance for EV applications.</li><li>• Worked on Battery Management Systems (BMS) and motor controllers, contributing to diagnostics, testing, and servicing of electric vehicle powertrains.</li></ul> |                   |
| <b>HVAC Testing Intern - Daikin Air Conditioners, India</b>                                                                                                                                                                                                                                                                                                                                 | 03/2025 – 03/2025 |
| <ul style="list-style-type: none"><li>• Worked as an HVAC Testing Intern at Daikin Air Conditioners, India, focusing on home and commercial HVAC units. Assisted in performance testing, quality checks, and diagnostics of HVAC systems. Gained valuable hands-on experience in thermal evaluation and airflow optimization.</li></ul>                                                     |                   |
| <b>Maintenance Intern - Neyveli Legnite Corporation India Ltd</b>                                                                                                                                                                                                                                                                                                                           | 11/2023 – 12/2023 |
| <ul style="list-style-type: none"><li>• Acquired practical exposure to large-scale thermal power plant operations, including lignite mining, power generation, and energy distribution processes.</li><li>• Assisted in monitoring, maintenance, and performance analysis of industrial equipment, ensuring operational efficiency and safety compliance.</li></ul>                         |                   |

## PROJECTS

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| <b>Intelligent Devices to Improve The Commutating Sector</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 03/2025 – 08/2025 |
| <ul style="list-style-type: none"><li>• Developed an IoT-based vehicle safety system using ESP32 that integrates hardware and software to prevent vehicle ignition without helmet detection and alcohol-free verification.</li><li>• Designed and implemented real-time sensing and control mechanisms to enhance commuter safety, contributing to smart mobility solutions under the SIH problem statement.</li></ul> <b>Technology used:</b> ESP32 Microcontroller, Alcohol Detection Sensor (MQ-3), Helmet Detection Sensor, Arduino IDE, Relay Module.                                                                                                                                                                                                                                  |                   |
| <b>CNC Machine using CD Drivers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 01/2025 – 05/2025 |
| <ul style="list-style-type: none"><li>• Designed and developed a functional CNC machine using recycled CD drive components and advanced microcontroller-based control systems.</li><li>• Programmed precise motor control algorithms for coordinated three-axis movement, enabling accurate drawing and engraving operations.</li></ul> <b>Technology Used :</b> Arduino Uno, Arduino IDE, G-Code, Stepper Motor, Servo Motor, Motor Driver.                                                                                                                                                                                                                                                                                                                                                |                   |
| <b>DC Converter by using SMPS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 01/2024 – 01/2024 |
| <ul style="list-style-type: none"><li>• Designed and implemented a high-efficiency DC-DC converter using SMPS principles with Bluetooth-based remote control via a custom mobile interface and Arduino firmware, enabling efficient power regulation, seamless user-friendly interaction, and real-time performance monitoring.</li><li>• Integrated essential safety features such as overcurrent protection and thermal shutdown, focusing on real-time control, enhanced power and efficiency, and scalable IoT connectivity, while ensuring overall system reliability, robust fault tolerance, and long-term operational stability.</li></ul> <b>Technology Used :</b> SMPS, Buck Boost Converter, Dot Board, Autocad Electrical, Potentiometer, Ammeter, Voltmeter, Bluetooth Module. |                   |
| <b>Voltage Booster</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 08/2023 – 11/2023 |
| <ul style="list-style-type: none"><li>• Designed and implemented a DC-DC voltage booster circuit to efficiently step up voltage with stable output across varying input conditions.</li><li>• Optimized overall performance using inductors, capacitors, and MOSFETs, gaining hands-on experience in power electronics, circuit design, rigorous testing, while improving voltage stability, minimizing switching losses, and ensuring high efficiency under variable load conditions.</li></ul> <b>Technology Used :</b> Operational Amplifier, Capacitor, Resitors, Voltmeter, MOSFET, Some Fliter Circuits.                                                                                                                                                                              |                   |

## CERTIFICATES

- Introduction to Networks, CyberSecurity Essentials - Cisco
- UiPath Academy Automation Developer Associate Training - Uipath
- Completed over 15 courses spanning a diverse range of fields, including Full Stack Development, Machine Learning, Internet of Things (IoT), Programming, Power Systems, Solar Energy, and Power Generation.
- Energy Swaraj - Energy Literacy Training .

## AWARDS

### Electrical Innovation - SAE EBAJA

Awarded for outstanding innovation in the design and development of the electrical system for an all-terrain electric vehicle, demonstrating excellence in circuit design, efficient power management, system integration, and practical engineering application under real-world conditions.

### Overall Second in India - ATVC

Secured 2nd place at the national level in the ATVC competition, demonstrating excellence in vehicle design, performance, teamwork, innovation, advanced technical problem-solving, strategic planning, real-world application, and all-terrain mobility engineering under competitive conditions.

### Second Place in Nexathon - Chennai Institue of Technology

Achieved 2nd place in the Nexathon competition for presenting an innovative tech solution, showcasing strong problem-solving, teamwork, technical development, creative thinking, communication, adaptability, and efficient project execution skills under time-constrained conditions.

## SKILLS

**Matlab** — Circuit Analysising, Basic Programings | **AutoCad** — Circuit Desgining, Component Desgining and Testing | **Solid Works** — Circuit Designing, Measurements for the designed circuit | **Soft Skill** — Strong Communication, Problem Solving, Interdisciplinary Collaboration, Leadership | **Programing Languages** — Arduino Programing, Java, Python | **Networking Tools & Protocols** — UART, SPI, I2C, CAN, USB, HTTP/HTTPS, LoRA, WiFi